

μEC3 ESP32 C3 WROOM 02 Microcontroller Development Board

Product Overview

μEC3 is a high-performance, tiny microcontroller development board based on ESP32-C3 WROOM 02 module. The board runs on 32-bit RISC V single core processor with a maximum frequency of 160 MHz. The module has 4 MB flash memory and the board is specifically targeted for low power IOT sensing applications.

The module is equipped with a PCB antenna trace which facilitates 2.4 GHz WiFi and Bluetooth 5.0 (BLE) communication. Board programming is done via on-chip/native USB without using an external USB-UART bridge chip.

μEC3 is an embedded IOT high performance device that is highly suited for low power applications such as Smart homes, Industrial automation, Health care, Consumer electronics, Smart Agriculture, Low-power IoT Sensor Hubs, Low-power IoT Data Loggers etc.

With its high-performance module, the μEC3 board is compact and tiny (2.7 x 2.4 cm) unlike other dev boards which makes it highly suitable for space constrained IOT applications and wearables. μEC3 board supports firmware development on ESP-IDF, Arduino IDE, Micro Python/Circuit Python and other similar platforms.

Specifications

Board dimension	2.7 x 2.4 cm
Processor	32-bit single core RISC V microprocessor with max frequency 160 MHz
ROM/SRAM	384 KB ROM/400 KB SRAM on-chip
Flash Memory	4 MB
Wireless Communication	2.4 GHz Wi-Fi (802.11b/g/n), Bluetooth 5.0 (BLE)
IO Pins	14 GPIO pins, 5V power pin and 3.3V LDO output pin exposed.
Interface support	I2C, I2S, SPI, 12-bit ADC - 2 channels, PWM, UART
Built-in LEDs	GPIO6 for control/status LED. Separate LED for power indication
Antenna	On-board PCB antenna
USB Programming	USB-C port with native USB programming
Switches	On-board BOOT and RESET buttons